



Mercury levels in Danish Medieval human bones

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ABSTRACT

Mercury has been measured in Danish medieval bones interred at several cemeteries. Bone diagenesis is estimated to play an insignificant role based on measurements of soil samples taken vertically and horizontally away from the thighbone at three graves. Our study shows that mercury-containing medicine has been administered in 79% of the *leprosy* cases and 40% of the *syphilis* cases. A substantial number of the monks interred in the cloister walk of the Cistercian Abbey at Øm show heavily increased mercury levels, whereas none of the friars interred in the Franciscan Friary of Svendborg show elevated mercury concentrations. This is interpreted as originating either from exposure to mercury-containing red ink in the scriptorium of the Abbey, or perhaps slightly more likely, from preparing or administering mercury-containing medicine in the hospital of the Abbey.

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1. Introduction

Mercury was used throughout medieval times in Europe and therefore found its way into the body of most human beings living at that time. In the present study, we have investigated three groups of people potentially exposed to higher mercury levels than the average medieval man. The first two are victims of *leprosy* and *syphilis*. These diseases have pronounced expression on the skin, and it is known that diseases visible on the skin were treated with mercury-containing compounds, at least in late medieval times. The third group investigated consists of individuals interred in the cloister walks, presumably friars or monks who lived at the friary or monastery. A certain group of monks at an abbey had daily access to the scriptorium, either as readers or as scribes of illuminated manuscripts. It is well known that mercury sulphide was used throughout medieval Europe as a colour pigment for the bright red ink used extensively in illuminated manuscripts. Another group of monks were occupied with treating the sick, and in this capacity they could be exposed to higher levels of mercury from producing or administering the mercury-containing medicine.

As a control group for the *leprosy* and *syphilis* victims, we measured mercury levels in other individuals with no visible signs of disease on their bones from the same cemeteries. In the case of the monks and friars with possibility of access to the scriptorium or medicine making, a number of individuals from the adjacent lay cemetery were selected as a control group.

Our data from the control groups constitute a good estimate of the average mercury level in the non-exposed part of the medieval population. This estimate can act as basic information pertinent to the discussion of present day mercury pollution.

Mercury can be accumulated in human bones, but the mechanism is not yet clear. Presumably the Hg^{2+} ion substitutes for Ca^{2+} in the bone carbonate or hydroxyapatite (Geneser, 2004) although it is also possible that other species of mercury, e.g. $\text{Hg}(\text{CH}_3)_2$ could be present in the organic parts of the bone. It is considered a general fact that the concentration in bones is lower than in many other tissues of the human body. This is supported by Garcia et al. (2001), who found concentrations of Hg in human bone remains from Spain to be lower than 0.05 $\mu\text{g/g}$, whereas they found levels of 0.25 and 0.14 $\mu\text{g/g}$ in kidney and liver, respectively.

Very few studies have reported Hg levels in exhumed human bones. Yamada et al. (1995) measured the Hg concentrations in Japanese individuals from two different burial sites and from two different time periods (6–7th century and 12–17th century). They found concentrations ranging from 175 to 1700 $\mu\text{g/g}$ in a few cases.

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They also measured the concentration of Hg in soil from the graves. In most cases the Hg concentration in the soil was below the detection limit. When this was not the case, there was no correlation between soil and bone concentrations.

Diagenesis is an important issue to be addressed when analysing excavated bones. Diagenesis may change the trace element concentrations in the bones over the long times of interment due to processes such as leaching to the surrounding soil or groundwater, or uptake from the surroundings (Hedges, 2002). Trace elements can be associated with the bones in different ways. Trace elements can form complexes with the organic components; they can be adsorbed onto the surface of the hydroxyapatite, or incorporated into the hydroxyapatite matrix (Elliott and Grime, 1993). Different methods can be used to investigate the degree of diagenesis. Hedges and Millard (1995) used the porosity of the bone as a measure of the dissolution of the hydroxyapatite crystals, while Nelson and Sauer (1984) analysed soil samples to rule out the possibility of migration of elements from the bone to the soil.

Martinez-Garcia et al. (2005) measured the content of copper, calcium, iron, zinc and lead in human bones from the Cartagena region. The bone samples were from different historical times and recent times. Their results showed an increased amount of elements in the bones through time. However, Millard (2006) question these results and point out the possibility of diagenesis in these bones.

In order to elucidate if Hg had entered or left the bones after interment in the present study, we measured earth samples at 1, 2, 5, 10 and 20 cm horizontally and vertically away from three individuals in a deserted Danish medieval cemetery at Tjærby Ødekirke near Randers (Ødekirke means deserted church).

2. Samples selected

The bone samples that were used in the present study are kept in museum collections, deposited at the Anatomical Department at University of Southern Denmark, at the Øm Kloster Museum at Silkeborg, and at the Anatomical Department, Panum Institute, at University of Copenhagen. Table 1 gives the sample numbers and details of location. The sampling locations are shown in Fig. 1.

The Franciscan Friary in Svendborg was founded in 1236 and situated close to the medieval shore line. It was dedicated to St Catherine. Two cemeteries were investigated here. One is the friar's interments in the cloister walk. This began in late medieval times, probably after 1400. The interments in the cloister walk must have ceased between 1532, when the Friary was donated to the town of Svendborg, and 1537, when mendicant orders were prohibited in Denmark. The other is the laymen cemetery situated to the west of the friary's church. Only a small part of the cemetery has been excavated, but it seems that the majority of the burials used here are from after 1500. The cemetery was probably used until the middle of the 17th century.

Svendborg Sct. Nicolai is a parish churchyard with burials from ca. 1100 until 1800, situated a few hundred meters away from the Franciscan Friary in Svendborg.

The cemetery of the Franciscan Friary in Odense functioned as a mendicant churchyard where in the Middle Ages interment could be obtained by paying. After the Reformation it became a parish churchyard for a poor quarter of Odense, and remained active until the beginning of 19th century. So here we undoubtedly have a majority of post-Middle Age graves.

At the Cistercian Abbey at Øm we collected bone material from graves in both the cloister walk and the laymen cemetery north of the abbey. The laymen cemetery was perhaps started shortly after the foundation of the abbey in 1172. There were no parish rights in connection with Øm, so the interred deliberately wished a burial with the Cistercians, depending on their prayers to shorten the stay

in the purgatory. In light of the Lutheran movement, there have probably only been post-reformational burials in exceptional cases.

The abandoned church in Tjærby near Randers was a parish church. It is mentioned for the last time in 1524, and it was probably abandoned shortly after the Reformation. The church was undoubtedly built in the 12th century, so the burials can encompass ca. 400 years.

3. Diseases visible on the bone material

Three quite different diseases are considered here, leprosy, syphilis and the newly discovered and so far unpublished pathological condition tentatively called 'focal osteolytic syndrome' (FOS). The only thing these diseases have in common is that they left specific lesions on the bone and that they were present in Medieval and/or Early Modern Denmark. They were by no means the only diseases that might have led to mercury treatment of skin lesions; but they are the conditions for which we have osteological evidence. The two well-known diseases, leprosy and syphilis, are known to cause skin lesions; it is still unknown if FOS also caused skin lesions.

Leprosy is the most extensively studied and best understood of these bone diagnosed diseases. Following Boldsen (2005) and Boldsen and Freund (2006) the likelihood that an excavated skeleton belonged to an individual who suffered from leprosy can be determined by scores for seven lesions: (1) the remodelling of the edge of the nasal aperture; (2) atrophy of the anterior nasal spine; (3) atrophy of the alveolar process on the pre-maxilla; (4) porosity or perforation of the palate; (5) subperiosteal exostoses on the fibula; (6) general hypertrophy (previously called general swelling or porotic hyperostosis) of the fibula; and (7) changes to (the lower/palmar face of) the fifth metatarsal. The presence of a leprosy indicating lesion is scored '1' and the definite absence of one is scored '0'. These scores through a kind of weighted sum are used to create a summary statistical parameter λ . Positive values of this statistical parameter indicate leprosy and negative values indicate absence of leprosy. In this study all skeletons assumed to come from sufferers of leprosy have λ values over +1 and all controls have λ values under -1. Leprosy was a common disease in rural Denmark throughout the Middle Ages. In major towns and cities (e.g. Malmö and Odense) it appears that the disease was eliminated during the first half of the 14th century (Boldsen and Møllerup, 2006).

Contrary to leprosy, syphilis is primarily a disease of the Early Modern and Modern Period in Europe. The epidemiology of the paleopathological diagnosis of syphilis has not been studied in the same way as has the leprosy diagnosis. As a consequence of this, the sensitivity and specificity of specific syphilis-related lesions has not been studied. Following standard text books (e.g. Aufderheide and Rodriguez-Martin, 1998) the skeletons have been scored for seven syphilis indicating lesions: (1) radiating scars to the exterior of the braincase; (2) general hypertrophy¹ of the scapular end of the clavicle; (3) general hypertrophy of (the back side of) distal humerus; (4) general hypertrophy of proximal ulna; (5) general hypertrophy of distal radius; (6) general hypertrophy of distal femur; and (7) general hypertrophy and bending of the tibia (saber shin). The skeletons classified as cases in the present study all had at least two lesions indicating syphilis and those classified as controls had none.

Focal osteolytic syndrome (FOS) was first noticed during work on a series of skeletons excavated in 1996 from the early medieval cemetery in Nordby, just outside Århus in Eastern Jutland (Skov,

¹ Here the term general hypertrophy is used to describe the swelling of bone due to one of several pathological processes. Readers should be warned that these descriptions alone cannot be used as a valid scoring method for syphilis-associated lesions.

Table 1

Data for the samples analysed and results of the mercury determinations

Lab no.	Site	Sample ID	Description	Hg (ng/g)	Average Hg (ng/g)
KLR-6465	Od FF lay cemetery	GP90-G62	Femur, female 25–45 years, syphilis	140, 148, 146	145 ± 10
KLR-6466	Od FF lay cemetery	GP90-G241	Humerus, female, 30–40 years, syphilis, arm position B	19.4, 19.0, 13.7	17.4 ± 3.2
KLR-6467	Od FF lay cemetery	GP90-G269	Femur, male, 30–50 years, syphilis	56.5, 73.0, 56.0	61.9 ± 9.7
KLR-6468	Od FF lay cemetery	GP90-G28	Humerus, female, 25–35 years, syphilis	240, 385, 657, 378	415 ± 170
KLR-6469	Od FF lay cemetery	GBP82-G14	Femur, male, 30–45 years, syphilis	77.1, 113, 94.7	94.8 ± 18
KLR-6470	Od FF lay cemetery	GP90-G4	Femur + humerus, male, 20–25 years, syphilis, arm position D	37.8, 44.9, 42.5	41.7 ± 2.9, (femur)
				47.0, 49.4, 45.3	47.2 ± 3.3, (humerus)
KLR-6472	Od FF lay cemetery	GP90-G205	Humerus, male, 20–25 years, syphilis, arm position B	1500, 1740, 2160	1800 ± 330
KLR-6473	Od FF lay cemetery	GP90-G13	Femur, female, 27–33 years, syphilis, arm position B	223, 234, 204	221 ± 15
KLR-6474	Od FF lay cemetery	GP90-G170	Femur + humerus, male, 25–30 years, syphilis, arm position D	61.2, 53.5, 56.5	57.1 ± 4.0, (femur)
				36.3, 21.2, 28.2	28.6 ± 7.6, (humerus)
KLR-6475	Od FF lay cemetery	GP90-G165	Femur, male, 35–45 years, syphilis	145, 127, 154, 135	140 ± 9.9
KLR-6478	Od FF lay cemetery	GP90-G480	Femur + humerus, male, 35–45 years, syphilis	41.7, 49.0, 43.6	44.8 ± 3.1, (femur)
				61.4, 46.8, 47.3	51.9 ± 8.3, (humerus)
KLR-6479	Od FF lay cemetery	GP90-G341	Femur, male, 35–50 years, n.d.d.	97.4, 98.7	98.0 ± 8.3
KLR-6480	Od FF lay cemetery	GP90-G25	Femur, female, 25–35 years, n.d.d.	67.3, 68.8, 69.2	68.4 ± 4.7
KLR-6482	Od FF lay cemetery	GP90-G121	Femur, female, 35–45 years, leprosy	355, 318, 353	342 ± 24
KLR-6483	Od FF lay cemetery	GP90-G155	Femur, male, 30–40 years, n.d.d.	17.2, 17.4, 15.7	16.8 ± 1.2
KLR-6484	Od FF lay cemetery	GP90-G17	Femur, female, 25–32 years, n.d.d.	69.0, 62.3, 66.8	66.0 ± 4.6
KLR-6485	Od FF lay cemetery	GP90-G77	Femur, female, 25–35 years, n.d.d.	21.5, 19.5, 17.1	19.4 ± 2.2
KLR-6486	Od FF lay cemetery	GP90-G225	Femur, male, 27–37 years, leprosy	319, 301, 312	311 ± 22
KLR-6487	Od FF lay cemetery	GP90-G421	Femur, male, 18–24 years, n.d.d.	34.8, 35.1, 33.4	34.4 ± 2.4
KLR-6488	Od FF lay cemetery	GP90-G419	Femur, male, 29–39 years, n.d.d.	115, 129, 118	121 ± 8.4
KLR-6489	Od FF lay cemetery	GP90-G423	Femur, male, 20–25 years, n.d.d.	50.2, 54.4, 53.7	52.8 ± 3.7
KLR-6490	Od FF lay cemetery	GP90-G452	Femur, female, 18–24 years, n.d.d.	18.8, 17.6, 17.6	18.0 ± 1.3
KLR-6491	Od FF lay cemetery	GP90-G491	Femur, male, 20–25 years, n.d.d.	43.3, 41.6, 45.2	43.4 ± 3.0
KLR-6492	Od FF lay cemetery	GP90-G356	Femur, male, 30–40 years, leprosy	447, 679, 525	551 ± 120
KLR-6493	Od FF lay cemetery	GP90-G230	Femur, female, 25–32 years, n.d.d.	70.5, 74.9, 73.7	73.0 ± 5.1
KLR-6536	Od FF lay cemetery	GP90-G171	Femur, female, 35–45 years, n.d.d.	43.2, 62.1, 69.4	58.2 ± 14
KLR-6537	Od FF lay cemetery	GP90-G146	Femur, male, 25–35 years, n.d.d.	43.9, 39.9, 41.4	41.8 ± 2.9
KLR-6538	Od FF lay cemetery	GP90-G124	Femur, female, 18–23 years, n.d.d.	293, 285, 282	287 ± 20
KLR-6539	Od FF lay cemetery	GP90-G180	Femur, female, 28–35 years, n.d.d.	78.5, 67.8, 79.9	75.4 ± 5.2
KLR-6540	Od FF lay cemetery	GP90-G27	Femur, female, 32–38 years, n.d.d.	47.6, 49.1, 40.3	45.6 ± 3.2
KLR-6541	Od FF lay cemetery	GP90-G92	Femur, female, 20–30 years, n.d.d.	21.8, 21.1, 20.6	21.2 ± 1.5
KLR-6557	Svb FF cloister walk	IIB10	Femur, male	28.8, 26.0, 23.5	26.1 ± 1.8
KLR-6558	Svb FF cloister walk	IIB11	Femur, male	3.75, 5.15, 3.53	4.14 ± 0.9
KLR-6559	Svb FF cloister walk	IIB12	Femur, male, 25–30 years	104, 38.4, 193, 192, 172	140 ± 68
KLR-6560	Svb FF cloister walk	IIIB7	Femur, male	10.1, 8.56, 10.1	9.60 ± 0.7
KLR-6561	Svb FF cloister walk	IIIB8	Femur, male	37.9, 27.9, 27.0, 27.2	30.0 ± 5.3
KLR-6562	Svb FF cloister walk	IIIB9	Femur, male	13.5, 10.3, 13.0	12.3 ± 1.7
KLR-6563	Svb FF cloister walk	IIIA4	Femur, male, 14–18 years	69.1, 75.3, 82.3	75.6 ± 5.3
KLR-6564	Svb FF cloister walk	IIIA3	Femur, male, 28–38 years	71.5, 71.1, 75.7	72.8 ± 5.0
KLR-6565	Svb FF cloister walk	IIIA11	Femur, male	11.7, 11.3, 21.2, 13.5	14.4 ± 4.7
KLR-6566	Svb FF cloister walk	IIIA12	Femur, male	13.6, 16.1, 16.6	15.4 ± 1.1
KLR-6567	Svb FF cloister walk	IIIA1	Femur, male, 20–25 years	43.1, 22.9, 25.6, 36.2	32.0 ± 9.4
KLR-6568	Svb FF cloister walk	IIIA10	Femur, male	19.3, 24.5, 18.9	20.9 ± 3.1
KLR-6569	Svb FF cloister walk	IIIA14	Femur, male	18.8, 17.0, 16.5	17.4 ± 1.2
KLR-6570	Svb FF lay cemetery	AG39	Humerus, male	29.9, 35.1, 33.5	32.8 ± 2.3
KLR-6571	Svb FF lay cemetery	AG33	Femur, female	19.8, 20.1, 20.0	20.0 ± 1.4
KLR-6572	Svb FF lay cemetery	AG31	Femur, female, 35–55 years, leprosy	27.3, 20.0, 22.3	23.2 ± 3.7
KLR-6573	Svb FF lay cemetery	AG4	Femur, ca. 11 years	162, 156, 151	157 ± 11
KLR-6574	Svb FF lay cemetery	AG18	Femur, male, 50 + years, cancer	36.7, 27.0, 28.2	30.6 ± 5.3
KLR-6575	Svb FF lay cemetery	AG3	Femur, female	74.9, 74.4, 73.7	74.4 ± 5.2
KLR-6576	Svb FF lay cemetery	AG56	Tibia, male	13.2, 16.9, 15.4	15.1 ± 1.9
KLR-6577	Svb FF lay cemetery	AG104	Femur, female, 20–35 years	311, 224, 307, 249	273 ± 43
KLR-6578	Svb FF lay cemetery	AG93	Femur, male, ca. 50 years	17.0, 15.8, 17.1	16.6 ± 1.2
KLR-6579	Svb FF lay cemetery	AG74	Femur	25.9, 23.5, 28.5	26.0 ± 1.8
KLR-6580	Svb FF lay cemetery	AG55	Femur, 0–7 years	112, 97.5, 134, 148, 127	124 ± 20
KLR-6581	Svb FF lay cemetery	AG54	Femur, 12–20 years	78.1, 69.9, 81.8	76.6 ± 5.3
KLR-5447	Svb Sct. N cemetery	A1	Female, 19–20 years, syphilis	2860, 2530, 3980	3120 ± 760
KLR-6509	Ø CA lay cemetery	8.L3	Femur, female, 20–30 years, leprosy, arm position C	221, 178, 184	194 ± 23
KLR-6510	Ø CA lay cemetery	XIII	Femur, male, 38.8 (18.0–75.6) years, leprosy, arm position B	21.9, 20.3, 20.8	21.0 ± 1.5
KLR-6511	Ø CA lay cemetery	8.K	Femur, female, 25.8 (19.3–38.5) years, leprosy, arm position A	169, 170, 137	159 ± 19
KLR-6512	Ø CA lay cemetery	4.M	Femur, male, 28.3 (–1–43.6) years, leprosy, arm position B	121, 106, 135	121 ± 8.4
KLR-6513	Ø CA lay cemetery	8.Æ.2	Femur, male, 25–30 years, leprosy, arm position A	70.9, 71.7, 121	87.9 ± 29
KLR-6514	Ø CA lay cemetery	8.T.3	Femur, male, 30–40 years, leprosy, arm position C	30.8, 51.9, 30.0	37.6 ± 12
KLR-6515	Ø CA lay cemetery	8.U.3	Femur, male, 29.1 (23.4–37.0) years, leprosy, arm position B	92.8, 81.9, 103	92.5 ± 6.4
KLR-6516	Ø CA lay cemetery	8.C.6	Femur, male, 40–60 years, leprosy	91.8, 66.4, 86.9	81.7 ± 17

(continued on next page)

Table 1 (continued)

Lab no.	Site	Sample ID	Description	Hg (ng/g)	Average Hg (ng/g)
KLR-6517	Ø CA church	7.L1	Femur, male, 57.0 (31.9–84.5) years, leprosy, arm position C	312, 248, 218	259 ± 48
KLR-6518	Ø CA lay cemetery	8.S.3	Femur, male, 50–65 years, leprosy, arm position B	238, 210, 229	226 ± 16
KLR-6519	Ø CA lay cemetery	4.K.3	Femur, male, 41.4 (28.7–64.6) years, leprosy, arm position C	63.7, 32.1, 28.4	41.4 ± 19
KLR-6520	Ø CA lay cemetery	8.F.1	Femur, male, 30.4 (20.9–44.1) years, n.d.d.	39.7, 42.8, 45.8	42.8 ± 3.0
KLR-6521	Ø CA lay cemetery	7.U.5	Femur, male, 19.1 (0–34.4) years, n.d.d.	73.2, 62.9, 63.3	66.5 ± 4.6
KLR-6522	Ø CA lay cemetery	5.K	Femur, female, 25.4 (20.8–31.9) years, n.d.d.	35.5, 36.1, 36.4	36.0 ± 2.5
KLR-6523	Ø CA lay cemetery	4.H	Femur, male, 44.9 (21.0–86.0) years, n.d.d.	5.95, 10.6, 5.98	7.51 ± 2.7
KLR-6524	Ø CA lay cemetery	5.O.5	Femur, male, 40.0 (27.7–62.6) years, n.d.d.	11.7, 13.0, 14.5	13.1 ± 0.9
KLR-6525	Ø CA lay cemetery	8.P.1	Femur, male, 35–60 years, n.d.d.	25.0, 22.8, 23.8	23.9 ± 1.7
KLR-6526	Ø CA lay cemetery	8.B.2	Femur, male, 26.2 (–1–42.8) years, n.d.d.	44.4, 44.3, 42.8	43.8 ± 3.0
KLR-6527	Ø CA lay cemetery	8.O.3	Femur, male, n.d.d.	60.2, 64.5, 61.1	62.0 ± 4.3
KLR-6528	Ø CA lay cemetery	4.A.5	Femur, male, 15–19 years, n.d.d.	46.8, 45.6, 44.8	45.7 ± 3.2
KLR-6529	Ø CA lay cemetery	5.H.5	Femur, 12–18 years, n.d.d.	48.4, 38.8, 38.7	42.0 ± 5.6
KLR-6530	Ø CA lay cemetery	9.E	Femur, male, 23.27 years, n.d.d.	28.0, 26.0, 60.0	38.0 ± 19
KLR-6531	Ø CA lay cemetery	7.A	Femur, male, 38.9 (28.1–59.44) years, n.d.d.	29.7, 29.5, 29.5	29.6 ± 2.1
KLR-6582	Ø CA cloister walk	A257	Femur, male, 30–40 years	30.1, 25.3, 26.4	27.3 ± 1.9
KLR-6583	Ø CA cloister walk	6.L	Femur, male	31.4, 33.6, 25.7	30.2 ± 4.0
KLR-6584	Ø CA cloister walk	6.K	Femur, male	57.3, 59.9, 56.7	58.0 ± 4.0
KLR-6585	Ø CA cloister walk	6.F	Femur, male	90.1, 123, 141, 130	121 ± 22
KLR-6586	Ø CA cloister walk	41.C	Femur, male, 40–50 years	401, 574, 455, 499	482 ± 73
KLR-6587	Ø CA cloister walk	GED	Femur, male, arm position B	9.53, 29.8, 8.26, 24.4, 3.69	15.1 ± 11
KLR-6588	Ø CA cloister walk	GBG	Femur, male, 40–60 years, arm position A	254, 259, 265	260 ± 18
KLR-6589	Ø CA cloister walk	GBE	Femur, male, 40–60 years	16.4, 30.7, 10.3, 15.5	18.2 ± 8.8
KLR-6590	Ø CA cloister walk	GBK	Femur, male, 40–60 years, arm position A	99.5, 63.2, 78.7, 69.0	77.6 ± 16
KLR-6591	Ø CA cloister walk	GCA	Femur, male, 40–60 years	18.2, 33.5, 16.5, 16.4	21.2 ± 8.3
KLR-6592	Ø CA cloister walk	GBC	Femur, male, 60 + years, arm position A	85.1, 74.1, 51.5, 62.9	68.4 ± 15
KLR-6593	Ø CA cloister walk	GEE	Femur, male, adult	310, 306, 275	297 ± 21
KLR-6594	Ø CA cloister walk	GBH	Femur, male, 40–60 years, arm position C	412, 285, 238, 256	298 ± 78
KLR-6595	Ø CA lay cemetery	8.V	Femur, male, ca. 75 years	24.4, 26.1, 32.6	27.7 ± 4.3
KLR-6596	Ø CA lay cemetery	8.E.2	Femur, male, 30–40 years, arm position A	43.2, 37.8, 33.6	38.2 ± 4.8
KLR-6597	Ø CA lay cemetery	7.K.4	Femur	102, 78.7, 154, 110	111 ± 32
KLR-6598	Ø CA lay cemetery	7.N.4	Femur	19.0, 16.0, 22.7	19.2 ± 3.3
KLR-6599	Ø CA lay cemetery	7.K.5	Tibia, male, ca. 22 years	42.1, 58.7, 46.6	49.1 ± 8.6
KLR-6600	Ø CA lay cemetery	8.D.1	Femur	24.5, 24.2, 19.6	22.8 ± 2.8
KLR-6601	Ø CA lay cemetery	8.L.1	Femur, male, 35–45 years	56.4, 59.1, 51.1	55.5 ± 3.9
KLR-6602	Ø CA lay cemetery	7.U.4	Femur	94.0, 88.6, 156, 107	111 ± 31
KLR-6603	Ø CA lay cemetery	7.O.4	Humerus	55.5, 43.6, 47.1	48.8 ± 6.1
KLR-6532	Ø CA lay cemetery	VIIIa	Femur, male, 36.3 (20.2–58.7) years, Morbus Nordby	40.8, 34.1, 33.5	36.1 ± 4.0
KLR-6535	Ø CA lay cemetery	5.A	Femur, male, 31.0 (24.5–40.1) years, Morbus Nordby	22.7, 35.6, 27.9	28.7 ± 6.5
KLR-6476	Nordby	FHM3970-CX	Femur, female, 25–35 years, Morbus Nordby	15.2, 14.5, 15.5	15.1 ± 1.1
KLR-6533	Nordby	FHM3970-ES	Femur, male, 33.5 (25.7–46.6) years, Morbus Nordby	22.8, 23.3, 23.8	23.3 ± 1.6
KLR-6534	Nordby	FHM3970-FF	Femur, male, 31.2 (23.7–43.9) years, Morbus Nordby	27.1, 26.0, 26.4	26.5 ± 1.8
KLR-6342-1	Tjærby Ødekirke Randers	A1416 A1388	Soil sample 1 cm from A1416, horizontally	13.1, 13.2, 13.0	13.1 ± 0.9
KLR-6342-2	Tjærby Ødekirke Randers		Soil sample 1 cm from A1416, vertically	19.0, 20.5, 18.9	19.5 ± 1.4
KLR-6343-1	Tjærby Ødekirke Randers		Soil sample 2 cm from A1416, horizontally	18.3, 21.1, 21.3, 27.7, 26.3, 17.9, 20.1	21.8 ± 3.8
KLR-6343-2	Tjærby Ødekirke Randers		Soil sample 2 cm from A1416, vertically	17.5, 16.0, 16.9	16.8 ± 1.2
KLR-6344-1	Tjærby Ødekirke Randers		Soil sample 5 cm from A1416, horizontally	11.0, 12.7, 13.1	12.3 ± 0.9
KLR-6344-2	Tjærby Ødekirke Randers		Soil sample 5 cm from A1416, vertically	13.7, 13.1, 15.3	14.0 ± 1.0
KLR-6345-1	Tjærby Ødekirke Randers		Soil sample 10 cm from A1416, horizontally	21.3, 20.9, 19.7	20.6 ± 1.4
KLR-6345-2	Tjærby Ødekirke Randers		Soil sample 10 cm from A1416, vertically	12.3, 14.2, 14.2	13.6 ± 0.9
KLR-6346-1	Tjærby Ødekirke Randers		Soil sample 20 cm from A1416, horizontally	13.9, 14.1, 13.3	13.8 ± 1.0
KLR-6346-2	Tjærby Ødekirke Randers		Soil sample 20 cm from A1416, vertically	29.8, 26.4, 23.8, 29.3	27.3 ± 1.7
KLR-6347	Tjærby Ødekirke Randers		Femur	28.8, 23.0, 24.8, 24.4	25.3 ± 1.5
KLR-6348	Tjærby Ødekirke Randers		Femur	57.9, 59.0, 63.1	60.0 ± 4.2
KLR-6349-1	Tjærby Ødekirke Randers		Soil sample 1 cm from A1388, horizontally	9.01, 8.90, 10.7	9.52 ± 0.7
KLR-6349-2	Tjærby Ødekirke Randers		Soil sample 1 cm from A1388, vertically	15.3, 19.3, 16.7, 18.7, 18.1	17.6 ± 1.0
KLR-6350-1	Tjærby Ødekirke Randers		Soil sample 2 cm from A1388, horizontally	16.0, 14.9, 16.7	15.8 ± 1.1
KLR-6350-2	Tjærby Ødekirke Randers		Soil sample 2 cm from A1388, vertically	21.9, 20.2, 17.6	19.9 ± 1.4
KLR-6351-1	Tjærby Ødekirke Randers		Soil sample 5 cm from A1388, horizontally	15.2, 21.6, 15.6, 15.6	17.0 ± 3.1
KLR-6351-2	Tjærby Ødekirke Randers		Soil sample 5 cm from A1388, vertically	17.4, 18.6, 25.1, 18.9	20.0 ± 3.4
KLR-6352-1	Tjærby Ødekirke Randers		Soil sample 10 cm from A1388, horizontally	15.5, 16.2, 17.7	16.5 ± 1.1
KLR-6352-2	Tjærby Ødekirke Randers		Soil sample 10 cm from A1388, vertically	11.8, 11.3, 11.6	11.6 ± 0.8
KLR-6353-1	Tjærby Ødekirke Randers		Soil sample 20 cm from A1388, horizontally	15.9, 17.4, 17.0	16.7 ± 1.2
KLR-6353-2	Tjærby Ødekirke Randers		Soil sample 20 cm from A1388, vertically	12.8, 17.2, 13.6, 13.9	14.4 ± 1.9
KLR-6355	Tjærby Ødekirke Randers	A1377	Femur	27.2, 23.8, 27.7, 22.5, 34.1, 27.2, 22.7, 29.9	26.9 ± 3.9
KLR-6356-1	Tjærby Ødekirke Randers	A1377	Soil sample 1 cm from A1377, horizontally	29.2, 35.0, 24.4, 22.2, 20.5	26.3 ± 5.9
KLR-6356-2	Tjærby Ødekirke Randers		Soil sample 1 cm from A1377, vertically	961, 1093, 1220, 951, 772	999 ± 170
KLR-6357-1	Tjærby Ødekirke Randers		Soil sample 2 cm from A1377, horizontally	313, 927, 665, 1104, 472	696 ± 320
KLR-6357-2	Tjærby Ødekirke Randers		Soil sample 2 cm from A1377, vertically	1877, 1261, 508, 689, 568, 1414	1050 ± 550
KLR-6358-1	Tjærby Ødekirke Randers		Soil sample 5 cm from A1377, horizontally	148, 478, 69.5, 97.90, 345, 75.0	202 ± 170
KLR-6358-2	Tjærby Ødekirke Randers		Soil sample 5 cm from A1377, vertically	334, 890, 894, 664, 409	638 ± 260
KLR-6359-1	Tjærby Ødekirke Randers		Soil sample 10 cm from A1377, horizontally	58.8, 52.4, 35.3	48.8 ± 12
KLR-6359-2	Tjærby Ødekirke Randers		Soil sample 10 cm from A1377, vertically	317, 339, 434, 479	392 ± 77
				28.6, 18.1, 28.8, 21.3, 21.8, 20.2	23.1 ± 4.5

Column 1, laboratory numbers; column 2, sample location (Od FF, Odense Franciscan Friary; SvB FF, Svendborg Franciscan Friary; SvB Sct. N, Svendborg Sct. Nicolai church; Ø CA, Øm Cistercian Abbey); column 3, sample ID; column 4, description, including gender and age determination, type of grave, and visible signs of diseases (n.d.d., no discernable disease; column 5, Hg concentrations determined in at least three independent runs; column 6, average Hg concentration and ± 1 standard deviation.



Fig. 1. Map of Denmark showing the positions of the cemeteries included in the present study. 1, Øm Cistercian Abbey; 2, Nordby; 3, Odense Franciscan Friary; 4, Svendborg Franciscan Friary.

2002). The condition consists of *intra vitam* erosions of compact bone surface and abscesses in trabecular bone. Usually the trabecular abscesses are connected to the compact bone erosions, but so far it is unknown if they occur independently of the bone surface lesions. Fig. 2 shows a skull from the Nordby series with numerous FOS lesions on the braincase, and Fig. 3 shows a closeup of one of these lesions.



Fig. 2. Lateral view of the skull of skeleton CX from the Nordby Early Medieval skeletal series. The numerous depressions are the typical signs of FOS. They all show vital reaction so they cannot be caused by post mortem damage.

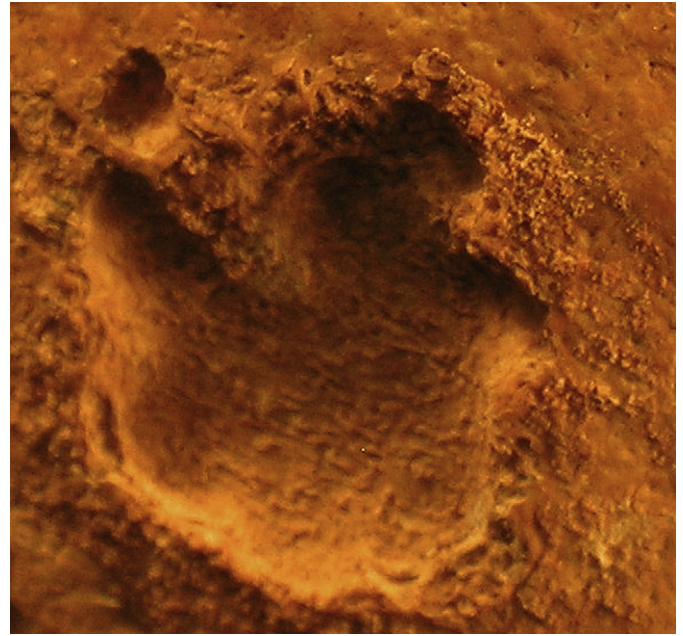


Fig. 3. Closeup of one of the lesions on the skull shown in Fig. 2 (skeleton CX from the Nordby series). This lesion is situated on the right parietal bone close to Lambda. The elevated rim of the lesion and the formation of new bone clearly illustrate the fact that this lesion was present at the time of death.

The FOS lesions can be found on all parts of the skeleton. Sometimes it can be difficult to distinguish between FOS lesions and *post mortem* damage to the bones. For this study, cases had at least two lesions showing clear evidence for vital reaction (as illustrated in Fig. 3). The controls were selected among well-preserved skeletons with no FOS lesions and with no *post mortem* damage that could be FOS lesions with no vital reaction.

4. Sample treatment

The skeletons stored at the University of Southern Denmark are mainly from medieval graves in the region of Southern Denmark. They have been excavated through the last ca. 40 years and brought to the University of Southern Denmark for storage. They are stored in cardboard or wooden boxes in a basement facility where examination of the bones is also carried out. The skeletons were screened for various diseases and selected according to the following criteria: a group of specimens with syphilis, another group of specimens with leprosy, and two groups of specimens with no discernable disease, the latter constituting the control groups.

The bone samples for Hg analysis were preferably taken from the thighbone (femur), whereas samples for radiocarbon dating were taken from the shinbone (tibia).

4.1. Sampling for Hg analysis

Because of the small amounts of Hg present in the samples great care was taken in order to avoid contamination. Anyone working with the bones was equipped with plastic gloves, hairnet, and a mask covering the mouth and nose as standard procedure in order to avoid contaminating the samples during the sampling procedure. The mask also prevented the person working with the bone from inhaling bone dust during sampling.

Sampling was carried out using a Dremel MultiPro® electric drill (model 395) with adjustable drilling velocity and equipped with a small 2 mm diameter drill. An ethanol flame was used to

evaporate any Hg present on the surface of the drill and other metal utensils prior to sampling.

Before the sampling, the working table was covered with a sheet of white A4 paper to prevent possible dust from the table contaminating the sample. A sheet of aluminium foil was placed on top of the white paper and another smaller sheet of Al-foil on top of that. The bone was placed on top of the small Al-foil and the drill was taken into the ethanol flame and left to cool a little. Then the surface of a bone area of approximately 2 cm² was drilled off to rid the bone of any surface contamination (see Fig. 4). The bone was brushed gently to remove any dust, and the upper layer of Al-foil was removed and discarded. After heating the drill again, the drilling of the sample material used for the analyses was done on top of a new clean sheet of Al-foil. Only compact bone tissue was used in our measurements. A sheet of white paper was used to transfer the bone dust to a sealed glass container, which was stored until the analysis took place. The upper layer of Al-foil was replaced before a new bone was sampled. Gloves were replaced together with the top layer of Al-foil, whereas the lower layer of Al-foil was replaced only for every third or fourth sample.

4.2. Sampling for radiocarbon dating

Using a hacksaw a V-shaped piece was removed from the tibia and placed in a small plastic bag. The bag was then sealed and mailed to the radiocarbon dating laboratory in Kiel for further pre-treatment. Prior to sampling, the saw was sterilized in an ethanol flame and gloves, hairnet, and mask were used during sampling. The samples dated at the Copenhagen radiocarbon laboratory were taken many years ago by various other researchers and submitted to the radiocarbon laboratory. Usually the thighbone (femur) was used by the Copenhagen laboratory, although not consistently.

5. Analytical methods

Mercury was analysed by atomic absorption in a dedicated analyser, an Advanced Mercury Analyser 254 (AMA 254). The machine was placed in a special laboratory, where great care was taken to avoid contamination with Hg. The personnel working in the lab were equipped with overall disposable body wear, hairnet, clean shoes and plastic gloves. The stainless steel utensils used for handling bone dust, soil or standard material were heated in an ethanol flame prior to use. The samples were weighed to an accuracy of ± 0.1 mg.

The AMA 254 machine made further sample treatment prior to analysis unnecessary. Samples were placed in a small container made of Ni, weighed and inserted into the AMA 254. In the machine, the sample was first heated to 120 °C in order to remove water. Then the temperature was raised to 850 °C in order to decompose the sample and release the Hg. During analysis, O₂ was used as a carrier gas and passed in a steady stream through the machine, partly to ensure a one-way flow of material and partly to ensure total combustion. The combusted gas was first carried to a catalyst tube that removed compounds containing halogens, nitrogen and nitrogen oxides and then past a gold mesh which trapped all Hg onto its surface. The rest of the combusted material left the machine with the carrier gas. The gold mesh was always kept at a temperature of at least 120 °C to prevent water from condensing. After the trapping was completed, the gold mesh was isolated. The trapped mercury was then released by heating the mesh to 950 °C for a short period of time after which the mesh was cooled rapidly in order to be ready for the next analysis. The evaporated Hg was carried into a measurement chamber consisting of two cells, a long and a short, where it was irradiated by a mercury lamp that emitted light with a wavelength of 253.65 nm. The light not absorbed by the mercury in the sample was registered by a detector placed behind a filter and the result was transmitted to a computer where the weight of the sample was entered. The result was given in both absorbance units and in ng/g. The machine could measure mercury over a scale of 0.5–600 ng Hg.

To ensure the optimal performance of the AMA a standard material was regularly analysed. The material was acquired from the National Institute of Standards and Technology (NIST) in USA and consists of apple leaves (SRM 1515) with a known concentration of mercury. Our particular batch of apple leaves had an Hg concentration of 44 ± 4 ng/g. If the average measurements of the apple leaves on a certain day were outside the interval 44 ± 4 ng/g, all the results for that day were adjusted by a factor forcing the standard material to exhibit a Hg concentration of 44 ng/g.

All samples were run at least in triplicate and Grubb's test (Miller and Miller, 2000) was carried out on especially diverse results. The uncertainty of the machine was estimated to be about $\pm 12\%$ (one standard deviation), which was used in calculating average and standard deviation for the results.

6. Radiocarbon dating methods

Twelve individuals have been radiocarbon dated. The dating has taken place over a period of several years. Samples previously dated were preferred in the selection process. The radiocarbon dating was performed at two different laboratories, the conventional gas-counting laboratory in Copenhagen, Denmark, and the AMS laboratory in Kiel, Germany.

The Copenhagen laboratory used a conventional gas proportional counter equipped with a guard gas counter. The sample consumption was about 100 g of bone material. After exterior cleaning, the carbonate and apatite fractions of the sample were dissolved in 0.2 M HCl and discarded. In order to remove humic acids, the demineralised bone material was treated in 0.1 M NaOH at room temperature for ca. 1 h followed by 0.1 M HCl in ca. 1 h. The last acidic step was conducted in order to remove any carbonate possibly extracted from the atmosphere during the previous alkaline step. The bone collagen was dissolved overnight as gelatine in demineralised water at 85 °C and pH 3. Only the dissolved collagen was transferred to a vial for evaporation of water; any non-dissolved components were allowed to settle overnight and were discarded. After drying, the collagen concentration was found by weighing the yield. The collagen was converted into CO₂ in a combustion chamber filled with pressurized oxygen. The resulting CO₂ was then cleaned for ²²²Rn, nitrogen oxides and halogens, which



Fig. 4. A bone after sampling has taken place. The excavated area is clearly visible.

could potentially affect the counting. A small subsample of the gas was extracted and measured for the stable isotope ratio, $\delta^{13}\text{C}$, in order to correct the date for any isotopic fractionation that could have taken place in ancient times or in the laboratory during pre-treatment.

At the Leibniz Labor für Altersbestimmung in Kiel, Germany, the collagen content was first calculated from the concentration of nitrogen, determined by a colorimetric method as nitrate in a ca. 2 mg subsample of bone material, and only the amount of bone needed for a good AMS sample (0.5–2 g of bone) was sampled after exterior cleaning of the bone. The bone sample was crushed to 0.5–2 mm pieces to facilitate demineralisation and first treated with acetone, then rinsed with demineralised water to degrease. The sample was subsequently treated in much the same way as in the Copenhagen laboratory: demineralised in ca. 1% HCl (0.3 M) until constant pH <1, washed with $\text{H}_2\text{O}_{\text{dem}}$ till pH >4, humic acids removed for 1 h at room temperature with 1% NaOH (0.25 M), again washing, extraction with 1% HCl (20 °C, 1 h), and finally washing till pH >4 (Groote et al., 2004). The bone collagen was dissolved overnight as gelatine in demineralised water at 85 °C and pH 3. The non-soluble fraction, which may contain particulate contaminants, was filtered on a 0.45 μm pore silver filter. The gelatine solution was freeze-dried, and the gelatine was combusted into CO_2 (Longin, 1971).

6.1. Reservoir correction

Individuals who received a sizable portion of their diet from the marine food chain will exhibit a “reservoir age” corresponding to the reduction in ^{14}C concentration caused by the intake of fish and shellfish living in ^{14}C depleted waters. Assuming a $\delta^{13}\text{C}$ of –19.8 for a purely terrestrial diet and –10.9 for a purely “marine” diet and a standard reservoir age of 410 years, the correction for reservoir effect can be calculated as:

$$K_{\text{reservoir}} = 410 \times \left(19.8 + \delta^{13}\text{C}_{\text{sample}} \right) / 8.9^{14}\text{C-years} \quad (1)$$

This formula was used to calculate marine reservoir correction on human bone samples by the Copenhagen Radiocarbon Laboratory during the last 10 years of the laboratory's existence (see e.g. Rasmussen, 2000, in press). This value is then subtracted from the conventional radiocarbon age prior to calibration. The calibration is performed using the 2004 curves and the Groningen Calib program version 5.0.1 (Reimer et al., 2004).

In the Middle Ages (the Catholic period) the Cistercians were in principle not allowed to eat meat from four-footed animals. This meant that when they obeyed the rule, they mainly ate fish. The Franciscans were probably more pragmatic, but as a main rule all medieval people were expected to fast ca. one-third of the year (although some people mostly cheated on this). Fish was not served every day, but quite often. Medieval people preferred freshwater fish to saltwater species, and abbey people most likely ate even more freshwater fish than the common population, who were forced to eat a majority of saltwater fish. Such an intake of freshwater fish can constitute a problem for the radiocarbon dating technique, because it is well known that freshwater fish can exhibit a large reservoir effect. In the inland lake Tissø on Sealand, Denmark, for example, an average freshwater reservoir effect of 1034 years has been reported (Fischer and Heinemeier, 2003), and in the interior of the Danish fjords an average reservoir effect of 634 years has been observed (Heier-Nielsen et al., 1995). These values are considerably higher than the open marine value of ca. 410 years normally used in Danish waters, a value on which Eq. (1) is also based. Moreover, the $\delta^{13}\text{C}$ values used as an indication of the fraction of marine carbon in the diet (less negative taken as more marine) become meaningless in this respect when freshwater fish

is involved. A freshwater reservoir effect, i.e. larger than 410 years, can probably be ruled out for the dates of individuals from Svendborg, as the town is situated on the coastline and does not have large freshwater systems in its vicinity. For the individuals from Øm, there could well be a potential problem, as Øm is situated near a large freshwater system and a fairly long distance from the sea. The freshwater system consists of Denmark's second largest lake, Mossø, which is connected to other large lakes via the Gudenå small river system. However, only one individual from Øm (8.K) exhibited a significant reservoir effect. The calibrated date for this sample may therefore be artificially too old. For the other samples from Øm, the reservoir correction is so small that the dates cannot be significantly affected, even if there was an increased freshwater reservoir effect.

After the Reformation, the lent largely disappeared, and there is nowhere near the same reservoir effect problem, although some special social circles maintained a preference for freshwater fish for a while after the Reformation in 1536.

7. Results

7.1. Radiocarbon dating

The radiocarbon dating results are given in Table 2. In addition to the collagen, the Kiel AMS laboratory also dated the ‘bone rest’ fraction, i.e. the non-dissolved organic part of the bone remaining on the filter after filtering the gelatinized collagen. The ‘bone rest’ fraction consists of non-gelatinizable bone organics, but may also comprise organic matter originating from other parts of the human body being decomposed or polymerized. In this case, the non-dissolved part poses no problem regarding the radiocarbon dating; collagen and bone rest give the same age. The insoluble fraction may, however, also stem from polymerized conservational fluids, lacquer applied to make a basis for writing museum numbers on, or particles of organic matter invading the porous bone structure many years after interment. This could lead to age differences between collagen and the rest.

The results of the ‘bone rest’ dating are given in Table 2. The age difference between the collagen and the ‘bone rest’ is not statistically significant for KIA-29656 and 29660. These samples were thus not significantly contaminated and the age is in this respect reliable. For KIA-29657 and 29658 the ‘bone rest’ is significantly older, indicating the presence of an older non-soluble contaminant, which could stem either from particles invading the bone from the environment of the human body or from polymerised conservational fluids derived from radiocarbon-void crude oil products. The problem arises if some of this fraction is included in the dated collagen sample. This could conceivably happen if, e.g. the particle sizes of the contaminant were smaller than 0.45 μm , or the settling was incomplete. However, as the filtering and settling of the collagen fraction, dissolved as gelatine, removes the non-soluble contaminants from the collagen and enriches them in the bone residue fraction, and the residue contained only 10% and 2% of the carbon recovered for KIA-29657 and 29658, respectively, one may expect the residue age to be much more influenced. In view of the observed age differences, the age of the collagen fraction should be reliable. It is, however, somewhat disturbing that it is precisely these two samples that have slightly less negative $\delta^{13}\text{C}$ values than the rest of the samples from Øm Friary. The $\delta^{13}\text{C}$ values were measured at –17.5 and –19.6‰ VPDB in KIA-29657 and 29658, respectively. A $\delta^{13}\text{C}$ value near –17.5‰ VPDB is, however, not unusual amongst Danish medieval bones, as devout Catholics traditionally ate fish in the lent period if not more often. We thus believe that the collagen date probably is trustworthy.

The Copenhagen laboratory did not attempt to assess the presence of contaminants in this way. However, the Copenhagen

Table 2
Radiocarbon age determinations

Lab no., sample ID	Radiocarbon lab no.	Radio-carbon age before 1950	$\delta^{13}\text{C}$ (‰ VPDB)	Collagen content (wt %) recovered	Marine food intake (%)	Reservoir corrected age before 1950	Calibrated date AD at $\pm 1\sigma$, reservoir corrected
KLR-5447, A1	KIA-20907-K	200 $\pm$ 20	−18.6 $\pm$ 0.1,	/10.5	13	145 $\pm$ 20	1679–1939 ^a
	KIA-20907-R	85 $\pm$ 25	−19.1 $\pm$ 0.1				
KLR-6570, AG 39 femur	K-3753	520 $\pm$ 65	−19.6 $\pm$ 0.1		2	510 $\pm$ 65	1321–1349, 1391–1448
KLR-6571, AG 33 rib	K-3754	410 $\pm$ 65	−21.3 $\pm$ 0.1		0	410 $\pm$ 65	1432–1521, 1577–1582, 1591–1621
KLR-6572, AG 31 rib	K-3755	300 $\pm$ 50	−18.0 $\pm$ 0.1		20	215 $\pm$ 50	1644–1683, 1736–1805, 1934–1951
KLR-6573, AG 4 (unknown bone)	K-3756	360 $\pm$ 65	−18.6 $\pm$ 0.1		13	305 $\pm$ 65	1493–1602, 1614–1649
KLR-6542, 8.S.3	KIA-29656-K	365 $\pm$ 25	−19.8 $\pm$ 0.1,	4.3/8.7	0	365 $\pm$ 25	1462–1519, 1593–1619
	KIA-29656-R	390 $\pm$ 30	−19.8 $\pm$ 0.1				
KLR-6543, 8.K	KIA-29657-K	795 $\pm$ 25	−17.5 $\pm$ 0.1,	8.8/6.7	26	690 $\pm$ 25	1277–1297, 1373–1377
	KIA-29657-R	875 $\pm$ 20	−20.5 $\pm$ 0.1				
KLR-6544, 7.L.1	KIA-29658-K	640 $\pm$ 30	−19.6 $\pm$ 0.1,	23.4/18	3	630 $\pm$ 30	1296–1318, 1352–1390
	KIA-29658-R	805 $\pm$ 35	−19.4 $\pm$ 0.1				
KLR-6545, 8.L.3	KIA-29659	310 $\pm$ 20	−20.7 $\pm$ 0.1	16.8/9.7	0	310 $\pm$ 20	1522–1574, 1585–1586, 1626–1642
KLR-6546, G356	KIA-29660-K	450 $\pm$ 20	−19.1 $\pm$ 0.1,	8.4/4.1	8	415 $\pm$ 20	1444–1466
	KIA-29660-R	525 $\pm$ 40	−21.1 $\pm$ 0.1				
KLR-6548, G121	KIA-29661	330 $\pm$ 20	−19.7 $\pm$ 0.2	11.8/14.9	1	325 $\pm$ 20	1517–1529, 1539–1595, 1618–1634
KLR-6547, G225	KIA-29662	505 $\pm$ 25	−19.7 $\pm$ 0.1	18.5/10.2	1	500 $\pm$ 25	1416–1435

Sample ID, when different from the sample used for Hg determination the bone is indicated; Radiocarbon laboratory ID no., conventional radiocarbon age in radiocarbon years before 1950 and 1 standard deviation; the $\delta^{13}\text{C} \pm 1$ standard deviation given in ‰ relative to the VPDB standard, the Copenhagen values were measured on CO_2 in a mass spectrometer (KIA values were measured via AMS and do not have the same accuracy as values measured on CO_2 in a mass spectrometer); the collagen content of the sample in wt % based on N_2 determination/collagen in extraction; the relative amount of marine food given in % calculated from $\delta^{13}\text{C}$ and Eq. (1); the radiocarbon age corrected for reservoir effect via Eq. (1); the calibrated date given at ± 1 standard deviation.

^a This date consists of many intervals; only the outer bounds are given here.

dates are in this respect equally reliable, because large amounts of collagen (ca. 100 times more than in Kiel) were extracted and only the dissolvable part of the collagen was used for dating.

7.2. Mercury determination

The results of the mercury analyses are given in Table 1. The relative standard deviation of the bone samples is generally less than 12%, and similar values apply for two of the sets of soil samples. For one set of soil samples (around individual A1377) the relative standard deviation is somewhat higher (ca. 36%), probably due to inhomogeneities in the soil. This is to be expected, because small rock fragments (sand or gravel size fraction) cannot be avoided altogether, and the inclusion (or omission) of such a small rock fragment in the sample prior to crushing will lower (or increase) the mercury level significantly by acting as a dilution.

The results for the measurements of the apple leaf standard material are shown in Fig. 5. The dashed lines represent ± 1 standard deviation on the mean value, which is adjusted to be 44 ng/g. It can be seen that all apple leaf determinations fall within ± 2 standard deviations. An unintended event happened during the end of the measuring period, from February 2006 onwards: a tube venting exhaust air from the machine to outside the laboratory fell off and this caused the background level of Hg in the laboratory to go up. This can be seen in Fig. 5 as a sudden systematic offset of ca. 5 ng/g in the apple leaf curve. This offset was corrected for as an offset, not as a factor, in the final data reported here. Consequently our data series are unaffected by this event. Furthermore, it is evident from the data and discussion below that the medieval bones have a general background level of ca. 50–100 ng/g, wherefore contamination or inconsistencies of a few ng/g are not pertinent to the interpretation of the results.

7.3. Bone diagenesis and transport of Hg

It is not known where mercury is situated in the bone, whether it is present as ionic Hg^{2+} substituting for Ca^{2+} in the apatite structure, or as $\text{Hg}(\text{CH}_3)_2$ imbedded, e.g. in the nervous cell tissue. The levels present, ng/g, are too low for any feasible mapping technique to reveal even the location of Hg. Therefore, it is very difficult to assess whether diagenesis has taken place or not. Even so, bone diagenesis constitutes a serious threat to any interpretation of trace element data in bones.

In order to test if any mobilization of Hg has taken place, we have measured the Hg concentration in soil samples at various distances from *in situ* bones. The samples were taken around three

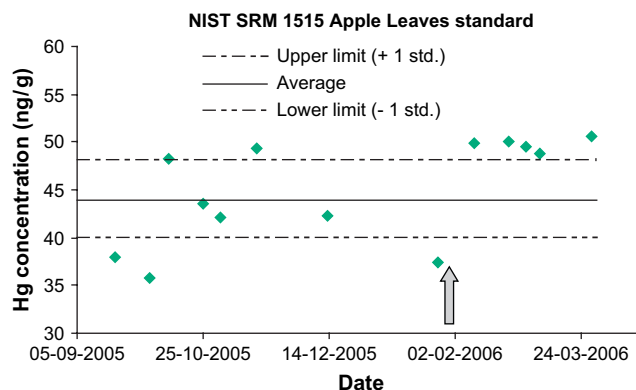


Fig. 5. Results of the analyses of the standard material, apple leaves from NIST standard SRM 1515. The dashed lines represent ± 1 standard deviation from the mean value of 44 ng/g (the specified concentration in the sample was 44 ± 4 ng/g). Our results were adjusted to give an average of 44 ng/g for the measurements for each day. The standard material was measured several times each day. The arrow indicates the time when the background level increased due to exhaust venting into the laboratory.

individuals at the deserted cemetery at Tjærby Ødekirke near Randers in Denmark. Samples were taken both vertically below and horizontally at distances of 1, 2, 5, 10, and if possible 20 cm from the femur surface. The results of the analyses are shown in Fig. 6. The points at distance 0 cm are measurements of the bone itself.

The three graphs of Fig. 6 look very different. In each case, however, the horizontally and vertically distributed samples show principally the same pattern. It should be noted that the samples at 1, 2, and possibly 5 cm are likely to reflect the Hg content of what was once the soft tissue, whereas the samples at 10 and 20 cm reflect the Hg content of the surrounding soil. The question is whether some Hg has been mobilized from the bone and/or the soft tissue and diffused into the soil, or whether Hg from the soil has invaded the tissue/bone.

In A1416, the bone is significantly higher in Hg than the tissue-affected soil and the more distant soil. In A1388, the situation is reversed: here the bone Hg value is only ca. half that of the soil samples. In A1377 the bone sample exhibits a very much lower Hg value (26 ng/g) than the extremely enriched close surroundings of the soft tissue (1100 ng/g). However, the most distant soil sample in the vertical direction is almost equal to the bone itself (24 ng/g), whereas the most distant soil sample in the horizontal direction is higher than the bone sample, but lower than the immediate surroundings (410 ng/g).

First, it should be noted that the graves in a medieval cemetery are spaced so close together that interference from neighbouring graves cannot be excluded. Sometimes undisturbed samples can only be sampled 10 cm, but not 20 cm away from the femur. This fact may introduce noise in our data. Second, there could conceivably be alternative Hg sources in the clothing or the coffin.

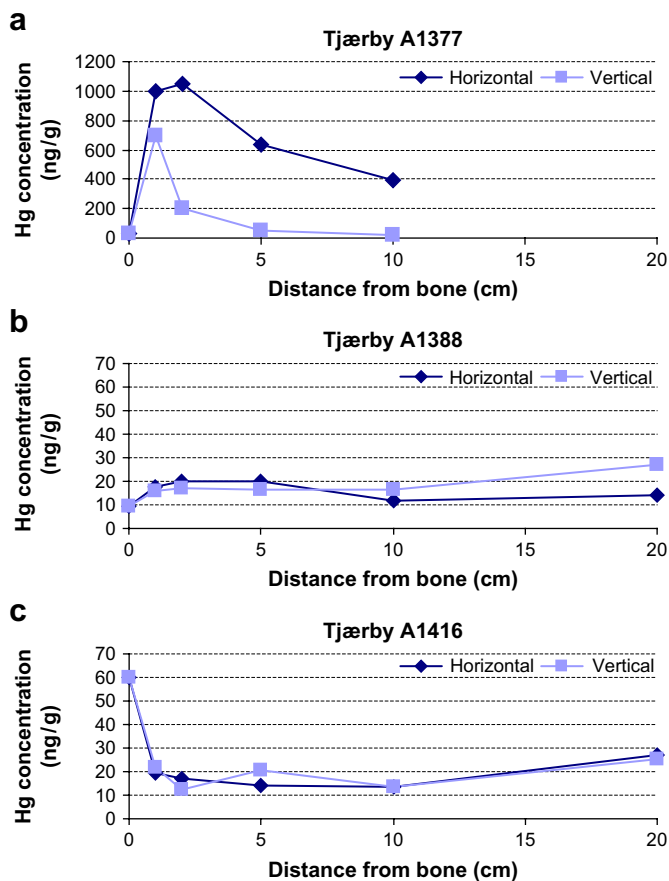


Fig. 6. Results of the mercury determinations in the soil samples taken at 1, 2, 5, 10, and 20 cm both horizontally and vertically (downwards) from the femur of three different individuals: a, A1377; b, A1388; c, A1416.

Having these caveats in mind, we will, however, suggest an interpretation of the soil data. A1416 could be an individual who had been exposed to large Hg levels earlier in his life, now reflected in high Hg value in the bone, and to lower Hg levels late in his life, now reflected in the low Hg values in the tissue. A1388 could be an individual who had experienced slightly lower Hg levels earlier in his life (slightly low bone value), and slightly higher Hg levels late in his life (higher tissue values). Alternatively, the Hg from the surrounding soil could have infiltrated the tissue-affected zone. For A1377 the interpretation could be that he had only been exposed to background levels earlier in his life, but at a time very close to his death, he experienced a large intake of Hg, possibly from the administration of Hg-containing medicine.

Whether or not these interpretations are indeed correct, it can be concluded from the three cases that no *systematic* transport has taken place with Hg leaving the bone material. Likewise, it can be excluded that systematic transport of Hg from the surrounding soil into the bones took place in this cemetery. This leads to the conclusion that we see no evidence for bone diagenesis related to mercury having taken place in this typical Danish medieval cemetery.

8. Discussion

8.1. Syphilis

Mercury administered as vapour – and most likely as skin tonic as well – was used in late medieval times as a cure for syphilis. Fig. 7 shows the mercury levels in the individuals in the control group and in the individuals with discernable signs of syphilis on the bone material, plus three individuals diagnosed to have FOS. For obvious reasons not all victims of syphilis did receive medical treatment, so it must be expected that a certain fraction of the individuals with syphilis symptoms exhibit only background levels of mercury.

From the data shown in Fig. 7a (the control group) we can deduce the average bone mercury concentration in individuals not exposed to unusually high levels of mercury. However, it is likely that a person not affected by syphilis could still be exposed to Hg, e.g. a monk tending the sick and administering Hg-containing medicine to others, a pharmacist producing the medicine or a novice preparing red ink for illuminating manuscripts. Likewise, it is possible that individuals in the control groups could have suffered from syphilis or other diseases that do not show up on the bone material, but on the other hand were severe enough to have prompted treatment with mercury-containing medicine. One of these alternatives seems to have been the case for G124, who is markedly higher in Hg than the rest of the control group. If we, therefore, exclude G124, the average Hg content of the rest of the control group becomes 53 ± 30 ng/g. Thus, judging from this number, in the data shown in Fig. 7b, six individuals show increased mercury levels high enough to suggest that they were treated with mercury-containing medicine (G13, G28, G62, G165, G205, and A1). The data from our limited study therefore suggests that $40 \pm 16\%$ ($=6/15 \times 100\%$) of the Danish population affected with syphilis severely enough to show up on the bone material received treatment with Hg-containing compounds.

8.2. Leprosy

Since it is well known that syphilis was treated with Hg-containing compounds in Late Medieval times, we thought it worthwhile to test if Hg was also used to treat other skin accentuated diseases such as leprosy, which was dominant in Early Medieval times. Fourteen samples from individuals with symptoms of leprosy were measured. The data of these and the control group are shown in Fig. 8.

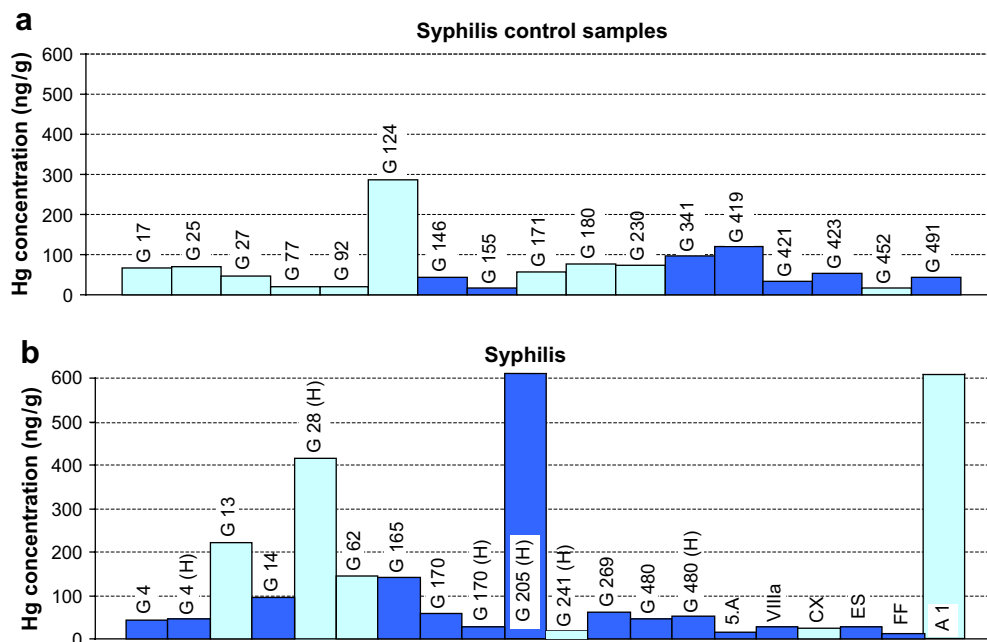


Fig. 7. (a) Results of the mercury determination of the control group showing no signs of syphilis from the laymen's cemetery at the Franciscan Friary at Odense. Light blue colour indicates females, dark blue males. (b) Results of the mercury determination of the group of individuals exhibiting clear signs of syphilis on their bones, plus five individuals having been diagnosed to have *Morbus Nordby* (5.A, VIIA, CX, ES, and FF). H in parentheses means that the sample was taken from humerus; all other samples were taken from femur. Light blue indicates females, dark blue males. The samples are from the Franciscan Friary at Odense (suffix G), the Cistercian Abbey at Øm (5.A and VIIA), the cemetery of Nordby (CX, ES, and FF), and finally the cemetery of Sct. Nicolai Church in Svendborg (A1).

All the individuals in the control group seem to be completely unaffected by Hg. The average Hg concentration in the control group is 34 ± 15 ng/g. Judging from this background level 11 out of the 14 individuals affected by leprosy were treated with Hg-containing medicine. This is as many as $79 \pm 24\%$. This high incidence of Hg-treated leprosy cases must be seen in the context that 11 of

these individuals were exhumed at the cemetery of the Cistercian Abbey of Øm, which is known to have been highly educated, also in the realm of medicine.

It seems that syphilis was introduced in Europe sometime shortly after America was discovered in AD 1492. At least we have never encountered any cases of syphilis that have been radiocarbon

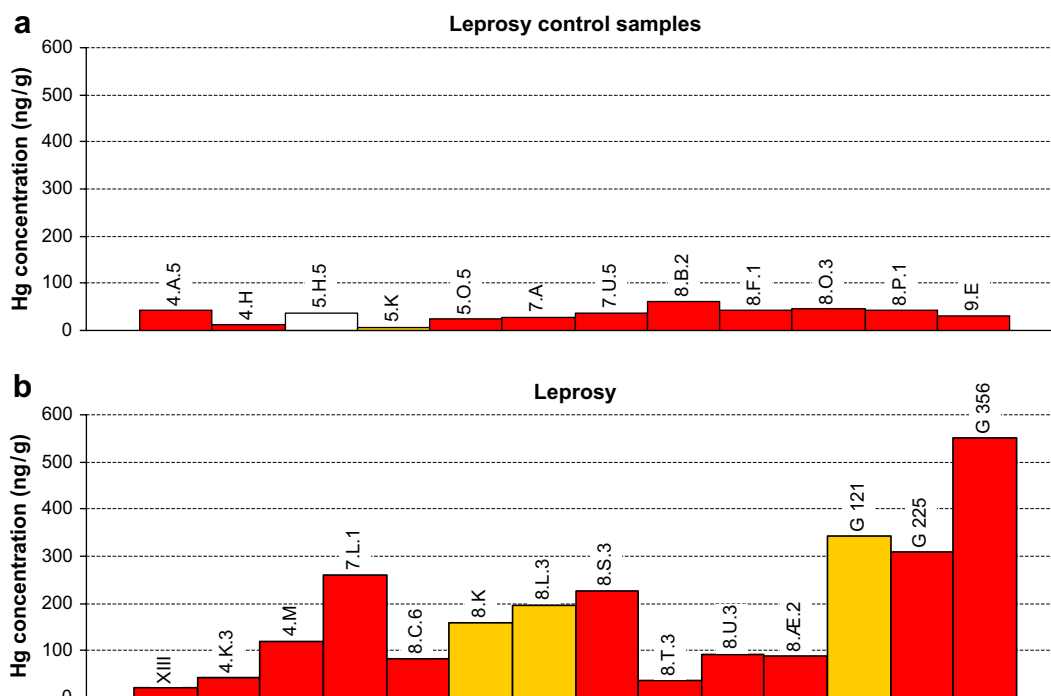


Fig. 8. (a) Results of the mercury determination of the control group showing no signs of leprosy from the laymen's cemetery at the Cistercian Abbey at Øm. Orange colour indicates females, red males, and white non-determined sex (a 12–18 year old juvenile). (b) Results of the mercury determination of the group of individuals showing signs of leprosy. The samples are from the Cistercian Abbey at Øm (the first 11) and from the Franciscan Friary at Odense (the last three). Orange colour indicates females, red males.

dated to a date significantly older than 1492. It also seems that the disease was very virulent right from the start and that syphilis took up much of the niche previously occupied by leprosy. In a very short time, syphilis became dominant and leprosy almost vanished. It was an aim of this study to test the assumption that syphilis took over more or less completely from leprosy at around 1500, i.e. can we identify any individuals showing certain signs of leprosy, which radiocarbon date to a time after 1492. The answer to this question is that two out of seven cases of leprosy treated with Hg-containing medicine are definitely after AD 1492 (G121 and 8.L.3). Therefore, our study shows that leprosy also thrived after the onset of syphilis, at least some time after AD 1500.

8.3. 'FOS': *Morbus Nordby*

Few diseases besides syphilis and leprosy can be revealed by the bone material, but *Morbus Nordby* or 'FOS' is one. Three samples exhibiting signs of FOS from Nordby Church were analysed (CX, ES, and FF), but none of them exhibited Hg levels in excess of the background. Therefore, from this limited data set it seems that FOS was not treated with mercury-containing medicine. This could be because the disease did not have a distinct expression on the skin of the victim. An alternative reason for the absence of excess Hg in these individuals could be that the community around the small Nordby Church was too poor for people to have access to mercury-containing medicine.

8.4. Access to the scriptorium or administration of medicine

The investigation of the friars exhumed in the cloister walk of Svendborg Franciscan Friary and monks from the cloister walk of the Cistercian Abbey in Øm reveal interesting results.

The data from the control group at the laymen cemetery at Svendborg Franciscan Friary show signs of elevated mercury levels

in three of the individuals (AG4, AG55, and AG104, see Fig. 9a). The average Hg concentration of the other nine individuals is 35 ± 24 ng/g. The data for burials in the cloister walk (Fig. 9b) show only one individual with elevated Hg concentration (IIB12). The average Hg value of the other 12 individuals from the cloister walk is 28 ± 23 ng/g, which is statistically identical to the level from the laymen cemetery. Thus in Svendborg Franciscan Friary, the friars are even less affected than the laymen. Only one individual showed an increased Hg level, whereas in the laymen cemetery in Svendborg three individuals showed somewhat increased levels of Hg.

In the control group at the Cistercian Abbey at Øm, consisting of ten individuals interred at the lay cemetery (Fig. 10a), two are slightly higher in Hg than the others, but the differences are so small, that we choose to interpret all of them as non-affected. The Hg concentration in the laymen cemetery at the Abbey of Øm is thus calculated to be 56 ± 33 ng/g. The data from the cloister walk burials at Øm show, however, significantly elevated mercury levels in five of the 13 monks included in this study (Fig. 10b). The Hg-affected individuals are 6.F, 6.K, GBC, GED, and GEE. Five individuals out of a total of 13 is $38 \pm 17\%$ ($=5/13 \times 100\%$). This is a surprisingly high number, and is in distinct contrast to the lack of Hg-affected friars in the cloister walk at the Svendborg Franciscan Friary.

There can be at least two explanations for the difference between Øm and Svendborg. One could be that precisely these five monks had long time access to the scriptorium in the Abbey of Øm, and ingested Hg from the red ink used in the scriptorium. The Hg could either be in the form of solid HgS, Hg⁰ vapour from heating HgS, or methyl-mercury from bacterial decomposition of HgS. An alternative explanation could be that the monks exhibiting high Hg values were active in the preparation of Hg-containing medicine in the hospital of the Abbey. In any event, the mercury introducing activity must have been an activity that took place at Øm and not in Svendborg.

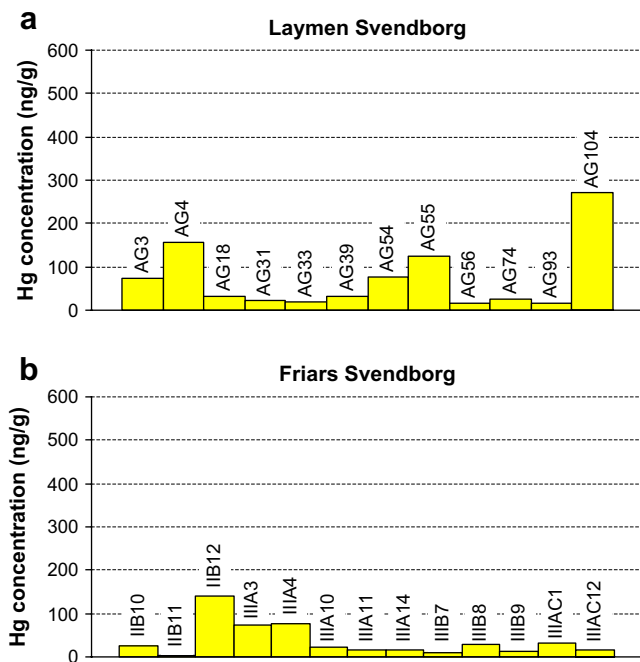


Fig. 9. (a) Results of the mercury determination of the control group from the laymen's cemetery at the Franciscan Friary at Svendborg. (b) Results of the mercury determination of the monks interred in the cloister walk at the Franciscan Friary at Svendborg.

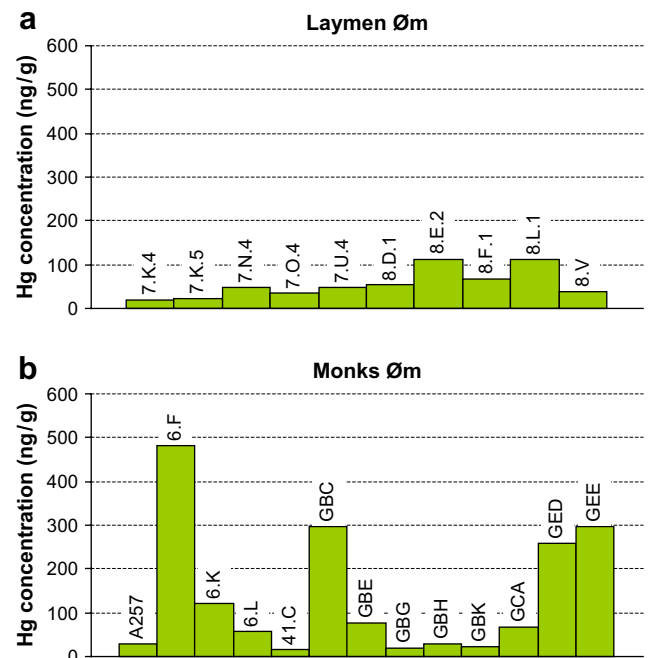


Fig. 10. (a) Results of the mercury determination of the control group from the laymen's cemetery at the Cistercian Abbey at Øm. (b) Results of the mercury determination of the monks interred in the cloister walk at the Cistercian Abbey at Øm.

Cistercians were not allowed to illuminate their incunabula with pictures, but they were allowed to illuminate starting letters, and they would therefore have used HgS extensively in their scriptorium. However, from our knowledge of these monastic orders in the Late Middle Ages (ca. AD 1400–1536), where most of the sampled individuals originate, one would probably expect a larger interest for literary activities at the Franciscans than at the Cistercians. In the High Middle Ages (ca. AD 1200–1400), it might have been different, but in the Late Middle Ages in Denmark the Franciscans were more intellectually active than the Cistercians. This fact could be taken as an indication that it is more likely that the high Hg levels in the five individuals at Øm are due to medicine preparation, rather than derived from activities relating to preparing and using ink in the scriptorium. However, this is not a definitive proof. Radiocarbon datings of more individuals may help to resolve this issue by placing the individuals in the either the High Middle Age or the Late Middle Age.

There are several factors relating to the dating of the bone material. In the period leading up to the Reformation in 1536, books were no longer written by hand but printed, and therefore the monastic use of red ink must have diminished. Although not statistically significant our data seems to indicate that the background level of Hg was higher in the control group for syphilis (53 ± 30 ng/g; see Fig. 7a), which is later than ca. 1500, than in the control group for leprosy (34 ± 15 ng/g, see Fig. 8a), which is generally earlier than ca. AD 1500. As we have not systematically radiocarbon dated the individuals exhibiting Hg background values, we do have in this sense – before and after 1500 – a dating system, which shows that the Hg levels seem to increase over time. This should be prioritized in future studies.

9. Conclusions

We have measured Hg concentrations in medieval bone material from six different cemeteries in Denmark and from soil samples in the vicinity of three individuals from one cemetery. Our data exhibit no signs of bone diagenesis.

Our study shows that Hg was indeed used for treatment of both syphilis and leprosy in Late Medieval times.

We see evidence of increased Hg levels in monks interred in the cloister walk in the Cistercian Abbey of Øm, but no evidence for increased Hg levels in friars buried in the cloister walk of the Franciscan Friary in Svendborg. This must reflect some differences in the use of Hg. We suggest two explanations: either handling and/or using red ink (HgS) in the scriptorium at Øm, or most likely the production or administration of Hg-containing medicine in the Abbey hospital at Øm.

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